

Exercise 2

Given a set of $n = 30$ observations, consider fitting the model $Y_i \sim \text{Bernoulli}(\pi_i)$ where $\text{logit}(\pi_i) = \beta_1 + \beta_2 x_i$, with x_i is a dummy variable taking value 1 for the first 10 observations and 0 otherwise. Fitting this model returns the following output

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.3863	0.5590	2.480	0.01314
x	-2.0794	0.7826	-2.657	0.00788

Null deviance	47.111
Residual deviance	39.112

$n = 30$
 $Y_i \sim \text{BERNOULLI}(\pi_i)$
 $\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_1 + \beta_2 x_i = \eta_i$
 $\eta_i = \beta_1 + \beta_2 x_i$

- a) Write the likelihood, log-likelihood and score functions for (β_1, β_2) . Write the fitted model.
- b) Compute the estimate of the probability $\hat{\pi}$ for $x = 0$ and $x = 1$. Obtain the odds for $x = 0$ and $x = 1$ and interpret them. Give an estimate of the odds ratio and interpret it.
- c) Test the hypothesis $H_0 : \beta_2 = -1$ vs $H_1 : \beta_2 < -1$.
- d) What are the two quantities "Null deviance" and "Residual deviance"?

LIKELIHOOD
 $L(\theta) = \prod \pi_i^{y_i} (1-\pi_i)^{1-y_i}$
 $\log L(\theta) = \log(L(\theta)) = \log\left(\prod \pi_i^{y_i} (1-\pi_i)^{1-y_i}\right) = \sum \log(\pi_i^{y_i} (1-\pi_i)^{1-y_i}) = \sum y_i \log \pi_i + (1-y_i) \log(1-\pi_i)$
 $= \sum y_i \tilde{x}_i^T \theta - y_i \log(1+e^{\tilde{x}_i^T \theta}) - (1-y_i) \log(1+e^{-\tilde{x}_i^T \theta}) = \sum y_i (\beta_1 + \beta_2 x_i) - \log(1+e^{\beta_1 + \beta_2 x_i})$

SCORE FUNCTION
 $\eta = \frac{d}{d\theta} \log L(\theta)$
 $\eta = \frac{d}{d\theta} \left[\sum y_i (\beta_1 + \beta_2 x_i) - \log(1+e^{\beta_1 + \beta_2 x_i}) \right] = \sum y_i - \frac{e^{\beta_1 + \beta_2 x_i}}{1+e^{\beta_1 + \beta_2 x_i}}$
 $\eta = \frac{d}{d\theta} \left[\sum y_i (\beta_1 + \beta_2 x_i) - \log(1+e^{\beta_1 + \beta_2 x_i}) \right] = \sum y_i x_i - \frac{e^{\beta_1 + \beta_2 x_i} \cdot x_i}{1+e^{\beta_1 + \beta_2 x_i}}$

FITTED MODEL
 $\log\left(\frac{\pi_i}{1-\pi_i}\right) = 1.3863 - 2.0794 x_i$

ODDS
 $\frac{\log(\frac{\pi_1}{1-\pi_1})}{\log(\frac{\pi_0}{1-\pi_0})} = \frac{\beta_1 + \beta_2}{\beta_1} = e^{\beta_2} \rightarrow e^{\hat{\beta}_2} = e^{-2.0794} = 0.125$
THE ODDS FOR $x=1$ ARE EQUAL TO THE ODDS OF $x=0$ MULTIPLIED BY 0.125

$\pi_i = 1 \rightarrow P(y_i=1|x_i=1) = \frac{e^{\beta_1 + \beta_2}}{1+e^{\beta_1 + \beta_2}}$
 $P(y_i=0|x_i=1) = 1 - \frac{e^{\beta_1 + \beta_2}}{1+e^{\beta_1 + \beta_2}} = \frac{1}{1+e^{\beta_1 + \beta_2}}$
 $\frac{P(y_i=1|x_i=1)}{P(y_i=0|x_i=1)} = \frac{e^{\beta_1 + \beta_2}}{1} = e^{\beta_1 + \beta_2} \rightarrow e^{\hat{\beta}_1 + \hat{\beta}_2} = 0.5$
WHEN $x=1$ THE ODDS OF FAILURE ARE MULTIPLIED BY 0.5 TO GET THE ODDS OF SUCCESS
WHEN $x=1$ I EXPECT 50 SUCCESS EVERY 100 FAILURE

$\pi_i = 0 \rightarrow P(y_i=1|x_i=0) = \frac{e^{\beta_1}}{1+e^{\beta_1}}$
 $P(y_i=0|x_i=0) = 1 - \frac{e^{\beta_1}}{1+e^{\beta_1}} = \frac{1}{1+e^{\beta_1}}$
 $\frac{P(y_i=1|x_i=0)}{P(y_i=0|x_i=0)} = \frac{e^{\beta_1}}{1} = e^{\beta_1} \rightarrow e^{\hat{\beta}_1} = 4$
WHEN $x=0$ THE ODDS OF FAILURE ARE MULTIPLIED BY 4 TO GET THE ODDS OF SUCCESS
WHEN $x=0$ I EXPECT 400 SUCCESS EVERY 100 FAILURE

ODDS RATIO
 $\frac{P(y_i=1|x_i=1)}{P(y_i=0|x_i=1)} = \frac{e^{\hat{\beta}_1 + \hat{\beta}_2}}{e^{\hat{\beta}_1}} = e^{\hat{\beta}_2} = 0.125$
THE ODDS OF GROUP $x_i=1$ ARE EQUAL TO THE ODDS OF $x_i=0$ MULTIPLIED BY 0.125

TEST
 $H_0: \beta_2 = -1$
 $H_1: \beta_2 \neq -1$
 $Z_{obs} \sim N(0,1)$
 $Z_{obs} = \frac{\hat{\beta}_2 - \beta_2}{0.7826} = -1.379$
 $\alpha^{obs} = 1 - \Phi(|Z_{obs}|) > 0.10$
SINCE $\alpha^{obs} < \alpha$ FOR $\alpha=0.1; \alpha=0.05; \alpha=0.01$ I REJECT
NO AT ALL CONFIDENCE LEVEL

DEV
 $D(MVLL) = 2 \{ \hat{l}_{(MVLL)} - \hat{l}_{(MODEL)} \}$
• MVLL MODEL $\rightarrow$ MODEL WITHOUT COVARIATES
• MODEL $\rightarrow$ MODEL WITH P COVARIATES
• SAT MODEL $\rightarrow$ MODEL WITH h COVARIATES
 $D(RES) = 2 \{ \hat{l}_{(SAT)} - \hat{l}_{(MODEL)} \}$